# Nmap: Network Recon & SYN Stealth Scan Architecture

How raw socket packet crafting, TCP half-open SYN scans, and firewall evasion techniques work at the Linux network kernel level.

## 1. TCP 3-Way Handshake vs SYN Scan

Standard connects complete SYN ➔ SYN-ACK ➔ ACK, logging in application access logs. Nmap SYN scan sends RST immediately after receiving SYN-ACK, tearing down the half-open socket with zero app-layer traces.

```
Client ➔ [SYN] ➔ Target:80
Target ➔ [SYN-ACK] ➔ Client (Port Open)
Client ➔ [RST] ➔ Target (Connection Aborted!)
```

## 2. Nmap Core Command Cheat Sheet

Essential flags for high-speed stealth reconnaissance across subnets:

```
# Stealth SYN scan all 65,535 ports at T4 speed
sudo nmap -sS -p- -T4 -v 192.168.1.1

# Aggressive service detection + default scripts
nmap -sV -sC -Pn -oA target_report 10.10.10.50
```

## 3. Firewall & IDS/IPS Evasion

Modern network firewalls track sequential port scans. Bypass signature detection using packet fragmentation and randomized decoy IPs:

```
# Fragment 8-byte MTU packets + 5 random decoys
sudo nmap -sS -f -D RND:5 -g 53 192.168.1.100
```

- `-f`: Splits TCP headers across packets
- `-g 53`: Spoofs DNS source port to bypass ACLs

## 4. NSE Scripts & Vuln Scanning

The Nmap Scripting Engine (NSE) automates CVE vulnerability checks, SSL cipher verification, and SMB enumeration:

```
# Scan for known critical CVE vulnerabilities
nmap --script vuln -p 80,443,445 10.10.10.100
```

- Check `/usr/share/nmap/scripts/`
- Custom Lua scripts for internal zero-days